

EPROM adapters for Commodore devices

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Many Commodore 8-bit devices employ ROMs to store their internal software. Over time the following ROMs, often produced inhouse by MOS Technology/CSG, were used:

- 2k x 8 bit: 2316
- 4k x 8 bit: 2332
- 8k x 8 bit: 2364
- 16k x 8 bit: 23128
- 32k x 8 bit: 23256

These devices followed the byte wide pin out originally thought out by Intel and even use Intel's basic device numbers. In case the contents of a ROM needs to be changed, about the only way this can be done is to replace it with an E(E)PROM. Here some puzzles arise.

One would assume that a 23XX ROM can simply be replaced by a corresponding 27XX EPROM, but this is not always the case. With Commodore equipment the following differences can be encountered:

- | | |
|---------------|--|
| 2316 <-> 2716 | Pin 21 is in many (if not all) cases an active high chip select pin on the 2316, but on the 2716, it is the Vpp pin, which must be tied to Vcc in normal applications. On some boards, Commodore has been using the active high chip select pin to implement a mechanism to disable the ROM (/NOROM signal). After replacing a 2316 with a 2716 directly, the active high chip select functionality on pin 21 is lost. |
| 2332 <-> 2732 | The main stumbling block here is the different location of the address pin A11 (pin 18 on the 2332, but pin 21 on the 2732). In addition, the 2332 has two chip select pins on pins 20 and 21. Their polarity is mask programmable, but in many cases pin 20 is active low and pin 21 active high, whereas the 2732 has two pins that can act as chip select pins (/CS on pin 18 and /OE on pin 20) and they are both active low. There is a solution to all this: the 2532 EPROM, which was produced by Texas Instruments and Hitachi. The only problem with this device is to find them nowadays and to get them programmed. The 2532 EPROM requires pin 21 to be logic high, so it is read compatible with the 2332 ROM with pin 20 active low and pin 21 active high. But: pin 21 on the 2332 ROM is in many (if not all) cases an active high chip select pin. Commodore has been using the active high chip select pin to disable the ROM (/NOROM signal) on some boards. In short: to provide a good path to replace a 2332 with a 2732 EPROM, an adapter is needed that provides the active high chip select on pin 21 and solves the pin out differences. |
| 2364 <-> 2764 | The 2364 is a 24 pin device with one chip select pin (which can be active low or active high), whereas the 2764 EPROM is a 28 pin device with the lower 24 pins having almost the same footprint as the 2364 ROM. The difference between the common footprint areas is the /OE input of the 2764 located on pin 22. With the 2364 ROM the corresponding pin is A12. A12 on the 2764 (and 27128) is located on pin 2, outside the 2364 footprint. Pin 18 of 2364 is the chip select, and can be active low or active high (Commodore used both versions). The four extra pins of the 2764 need to be connected to Vcc for normal operation, apart from pin 2. A good adapter needs to address the pin out differences, and provide an option for the chip select polarity. |

23128 <-> 27128 These devices have almost the same pin outs that make them usually interchangeable. No adapter is needed with Commodore equipment.

23256 <-> 27256 These devices have the similar pin outs and are usually interchangeable. Again no adapter is needed with Commodore equipment.

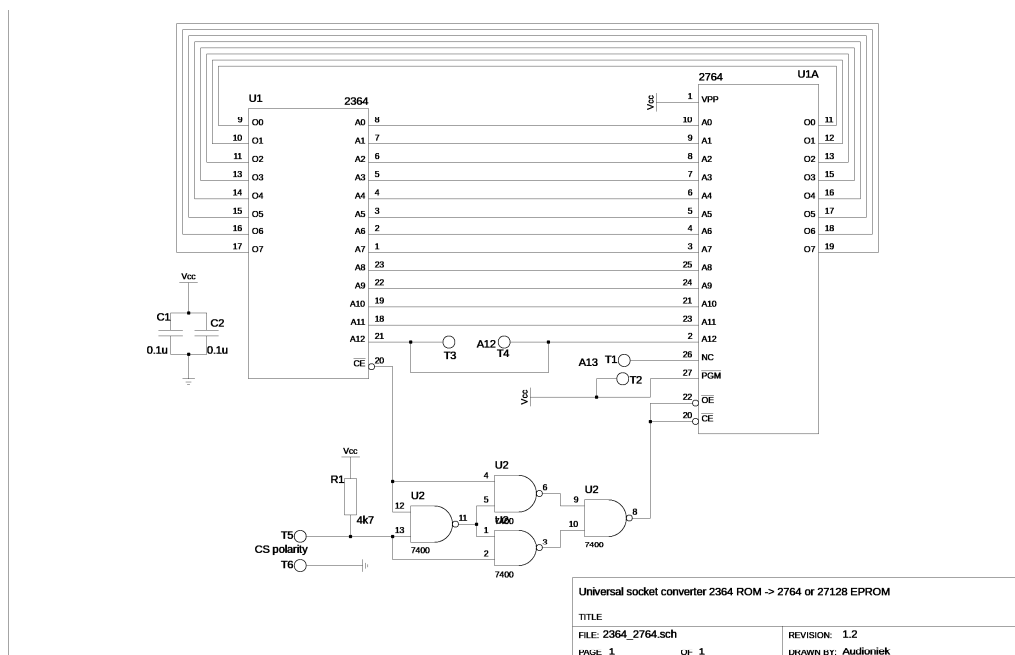
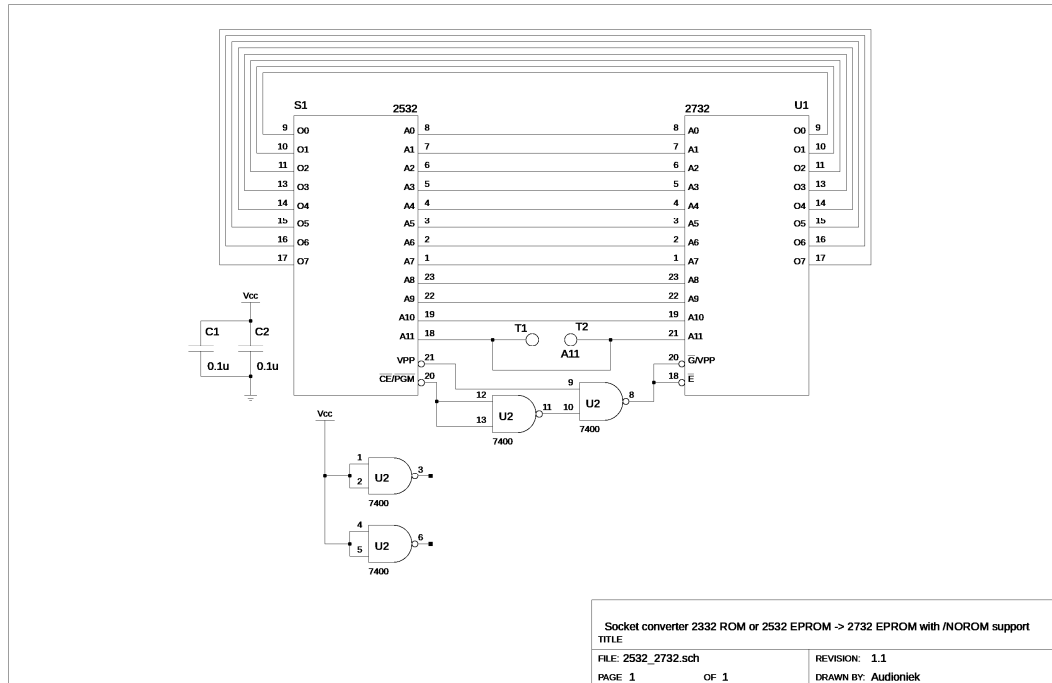
Final note: in practice all ROM types have one or more (usually all) chip select pin(s) that are mask programmable regarding polarity. The idea behind this is to simplify the design of the address decoding circuitry or in simple systems, maybe even do without it. In practice this was seldom made use of and most ROMs had polarities making them more or less compatible with the matching EPROMs. The cases described above do not cover all possible permutations, but do cover those encountered with 8-bit Commodore equipment in practice. If in doubt, consult a schematic diagram and the relevant data sheets.

Switching between two ROMs

Sometimes it is desirable to have the possibility to switch between two or even four ROMs that share the same socket. This is usually done by putting the contents of the two/four ROMs into the lower and upper quarters or halves of an EPROM having twice/four times the size and switching the most significant address line(s) of it to select the active ROM. As both the 2716 and 2732 as well as the 2764 and 27128 have compatible foot prints apart from the most significant address line(s), the adapter can be made usable for this purpose by providing an option for the address line(s) in question. To use a 2764 this way in a 2332 socket an adapter for 2364 <-> 2764 can be used with a minor modification. The same is valid for a 27128 in a 2364 socket.

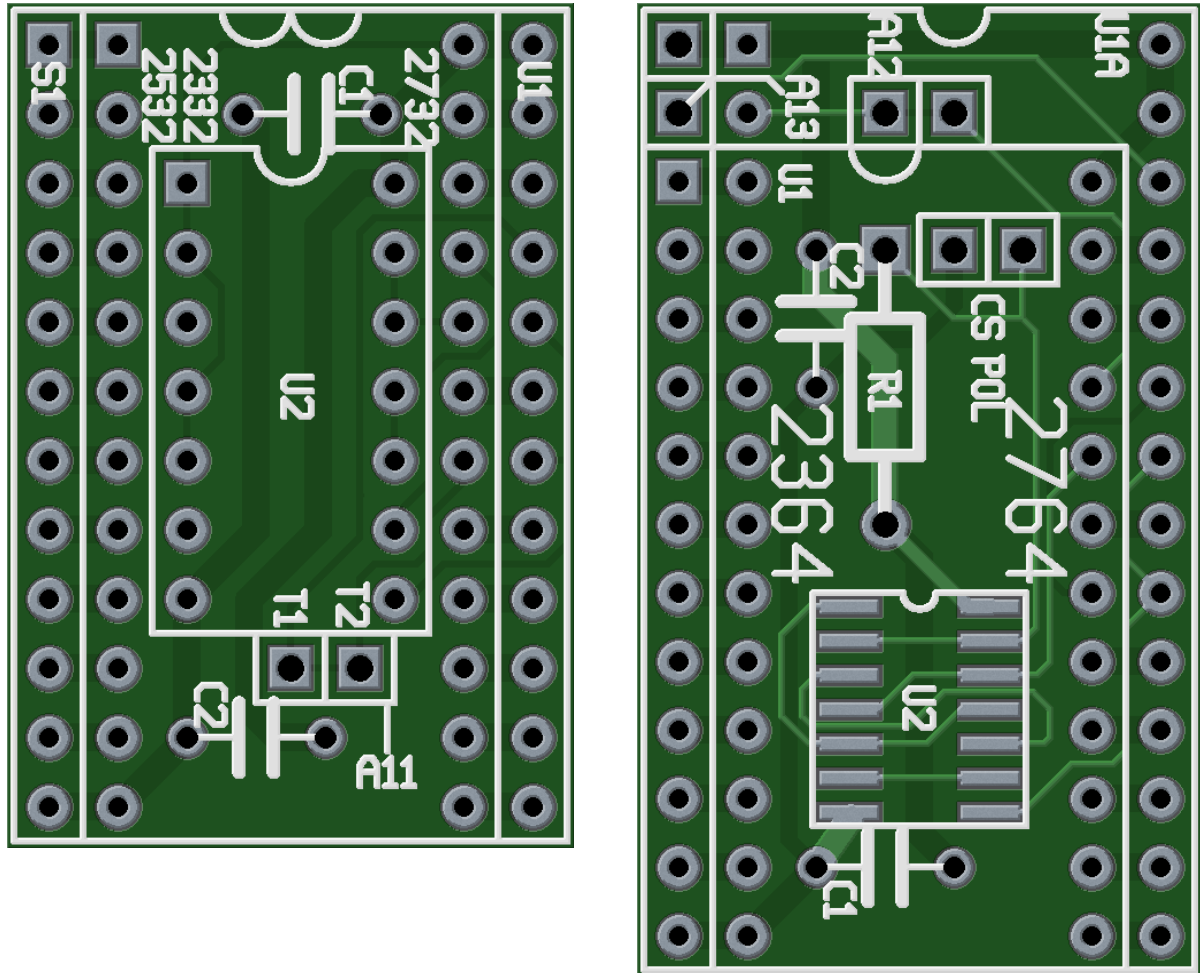
Schematic diagrams

Two adapters have been designed; one with a 2316/2332/2532 footprint, and one with a 2364 footprint. Here are their schematic diagrams.



The PCBs

For the two circuits presented before, two PCBs have been designed. Both are 0.1 inch wider than the footprint of the EPROM on top. This has been achieved by putting some components below the EPROM. Here are the component sides of the PCBs:



Circuit description

2332 footprint

The adapter has a 74LS00 built in, that cares for the active high chip select pin on pin 21 of a 2332 ROM as used in Commodore equipment. This is to provide the /NOROM function on later main boards of Commodore computers using 4 kbyte ROMs. The feature is compatible if the adapter is used in a 2316 foot print socket provided pin 18 is active low, pin 20 is active low and pin 21 is active high. As pin 18 is A11 on the 2332/2532, this pin can be made to take A11 out to provide the function 2x2316 to one 2732(A). For normal operation the hole A11 (T2) and the hole next to it are connected together with a small PCB trace on the bottom that can be easily cut. In order to provide polarity selection for one or both chip select pins the two unused gates of the 74LS00 are available. All traces to the inputs of the unused gates have been put in the bottom plane where they are accessible for additional wiring and/or trace removal/cutting; this may come in handy if for instance an extra inverter for the CS function(s) is needed.

2364 footprint

The adapter has an extra 74LS00 built in that is wired as an EXOR gate to provide a selectable inversion of the chip select pin 20 of the ROM. Due to space reasons this is an SMD device. Pin 26 of the EPROM socket is either brought out (it is A13 on 27128(A)) or can be connected to Vcc for normal read mode on a 2764(A) using a jumper. The path to A12 of the EPROM has also been made accessible. Normally it is connected to A12 of the 2364 through a small PCB trace on the bottom that can easily be cut if required, in case A12 is to be provided externally for the function 2x2332 in one 2764. As A13 on a 27128(A) is also accessible, the function 4x2332 in one 27128(A) is also available.

In practice this adapter is usable for the following cases:

2364 ROM -> 2764(A) EPROM: connect the hole for A13 with the hole next to it with a wire or jumper. A12 of the EPROM is already connected (with a PCB trace on the bottom) to A12 of the ROM, and pin 27 of the EPROM is tied to Vcc as required for read of a 2764(A). The device in the top socket is a 2764(A) EPROM. In case the chip select pin 20 of the 2364 is active low (in a 1541 floppy disk drive for example), the holes marked CS POL are connected or jumpered. Leave the pins CS POL unconnected for an active high chip select on ROM pin 20 (in an 8050 floppy disk drive for example).

2x 2332 ROM -> 2764(A) EPROM: The top device is a 2764(A) EPROM (so use the A13 jumper). A12 is now the select line and is connected outside the adapter. Cut the trace on the bottom side between the A12 hole and the hole next to it. Proviso/CAUTION: pin 21 of the socket the adapter input is now ignored. With Commodore equipment using 2332 ROMs this pin is either the /NOROM function (pulled up to Vcc) or tied/pulled up to Vcc anyway. Check before using the adapter is this fashion!

2x 2364 ROM -> 27128(A) EPROM: Select the chip select polarity of pin 20 of the 2364 ROM in the same way as described with the 2364 -> 2764(A) function. Leave the jumper for A13 off and connect the second pin (not the corner pin) of the jumper field to the ROM select line. The device in the top socket is a 27128(A) EPROM.

4x 2332 ROM -> 27128(A) EPROM: The top device is a 27128(A) EPROM (so leave the A13 jumper off). Cut the trace on the bottom side between the A12 hole and the hole next to it.

Proviso/CAUTION: pin 21 of the socket the adapter input is now ignored. With Commodore equipment using 2332 ROMs this pin is either the /NOROM function (pulled up to Vcc) or tied to Vcc anyway. Check before using the adapter is this fashion! The holes marked A12 and A13 select the 4 kbyte part of the EPROM to be used and are externally connected.

General notes

The two EPROM adapters do *not* provide a pull up for the address line(s) to be connected externally in case of switched ROMs in order to provide a default condition. Be sure to provide a defined logic level to these pins in all circumstances; do not leave them floating/not connected. The CS POL pin is pulled up on the 2364 adapter; the CS polarity therefore defaults to active high.

EPROM variants

The EPROM types involved (2716, 2732, 2764 and 27128) exist in multiple variants. The first is the inclusion of a C after the 27. These devices are built using CMOS technology, and can be used as an alternative. CMOS devices tend to use less energy than their NMOS counterparts. The second variant has an additional letter A at the end. This denotes a lower programming voltage which is of course important when programming these devices. After programming, all variants with the same basic model number are read compatible with each other and can be used with the adapters described here. When doubts arise, please consult the relevant datasheet(s).

The final essential point is access time. As these adapters will generally be used in equipment that use a 6502 compatible CPU running at 1 MHz, an access time of 450 ns is sufficient. In practice all EPROMs meet this value and are usually even faster nowadays.

Equivalent EEPROMs can be used in place of the EPROM types mentioned; 28(C)16 and 28(C)64's do exist. A 27128 can be replaced with a 28(C)256 with A14 tied to either GND or Vcc, as a 28(C)128 does not exist. Replacing a 2732 with an EEPROM is somewhat difficult: a 28(C)32 does not exist and when using a half of a 28(C)64 you run into the problem of a difference in pin count. In all cases, the EEPROM needs to be programmed/written beforehand; the adapters do not provide a path for the WR pin.

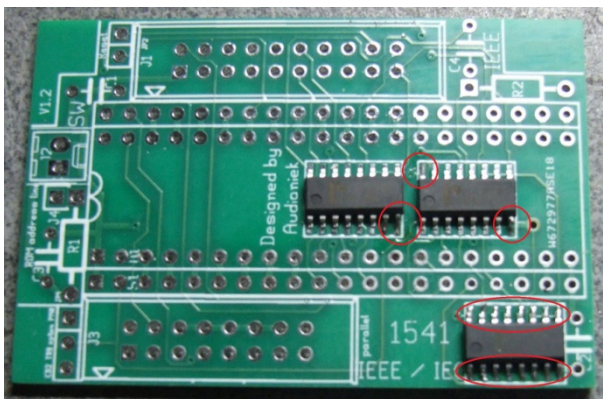
Building an adapter

An adapter is built the easiest in the following order:

- In case you build the 2364 footprint adapter with the SMD 74LS00, start with this device. If you have access to SMD soldering equipment, things should be straightforward. If not, the part can be mounted using conventional soldering by carefully doing the following. First add some solder to one of the PCB islands for the corner pins of the 74LS00 using a soldering iron with a very small tip and thin soldering wire. Remove the solder wire and the iron as soon as the solder starts to flow. The idea behind this is to leave as much flux in the solder on the PCB as possible.

Next put the 74LS00 in position and heat the pin covering the tinned PCB island and the PCB island itself simultaneously with the soldering iron. After the solder has flowed, the 74LS00 is now connected by one pin to the PCB. Now correct the position of the 74LS00 if required by aligning the pin diagonally opposite the already soldered pin with the PCB island for it, using the soldered pin as a hinge. If the position is OK solder the diagonally pin to its PCB island, again by simultaneously heating the pin on the part and the PCB island intended for it. This will establish the correct positioning of the part in the two significant directions.

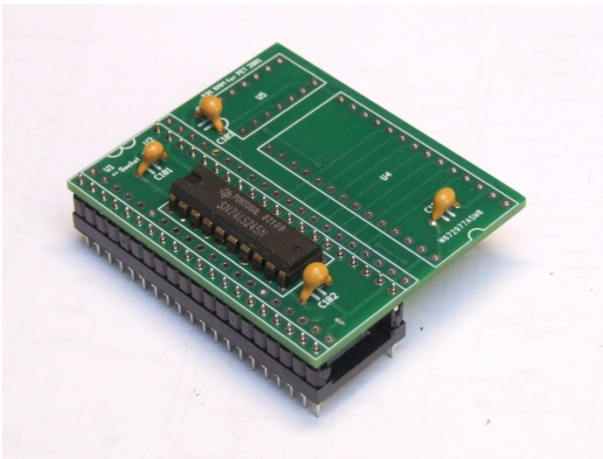
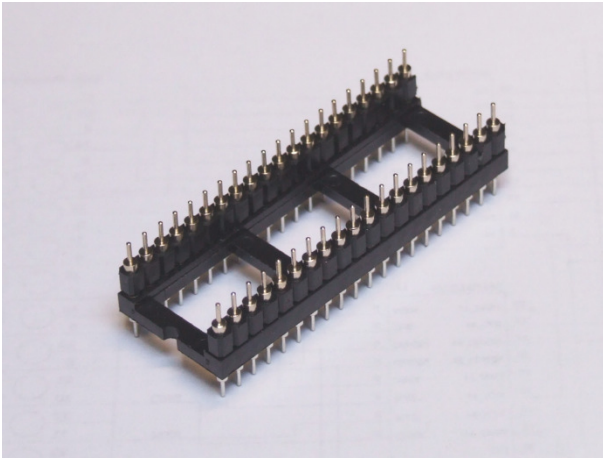
Solder the remaining pins of the 74LS00 to finish off:



The photo above illustrates the three steps from left to right on a different PCB;

- Mount all remaining components on the component side of the PCB, except the IC socket for the top EPROM. For the 2332 footprint adapter this includes the 74LS00;

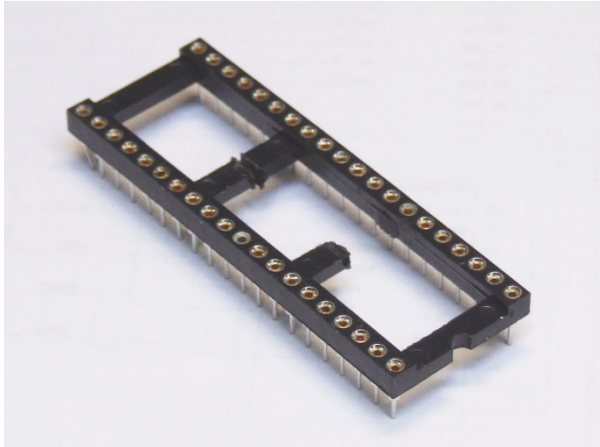
- Use the EPROM socket as a template by inserting the thin pins of the pin strips into it and insert the thick end of the pins into the PCB holes intended for them.



(Note: 40 pin socket and different PCB shown, but you will get the idea.);

- Solder the thick ends of the strips on the component side of the PCB.
- Carefully remove the socket for the EPROM from the pin strips;
- The 2364 footprint adapter has two jumper fields where the use of jumper pins is recommended: one for A13 and one for the CS polarity; the CS POL pins need to be angled. If you need the A12 pin, use an angled jumper pin for that field as well. The 2332 footprint adapter only needs angled jumper pins for A11 in case it is used for the function 2x2716->2732;

- Depending of the particular socket for the EPROM to be used, the bridges in the middle of it may interfere with parts already on the component side. Check if any need to be removed. Do not simply cut the bridge out of the socket with cutting pliers as this will probably snap the socket in two. Cutting pliers can be used directly when the socket is first put in a vice to prevent it splitting while cutting. As an alternative, heat a knife with a flame or your soldering iron and use the heated knife to "cut" the bridge in half. Then cut the bridge halves out using cutting pliers:

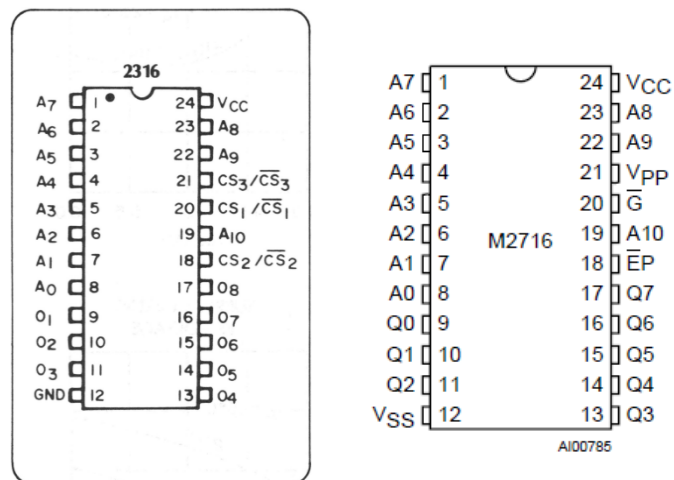


When using the A11 angled jumper pin on the 2532 footprint adapter, the bottom bridge needs to be removed as well;

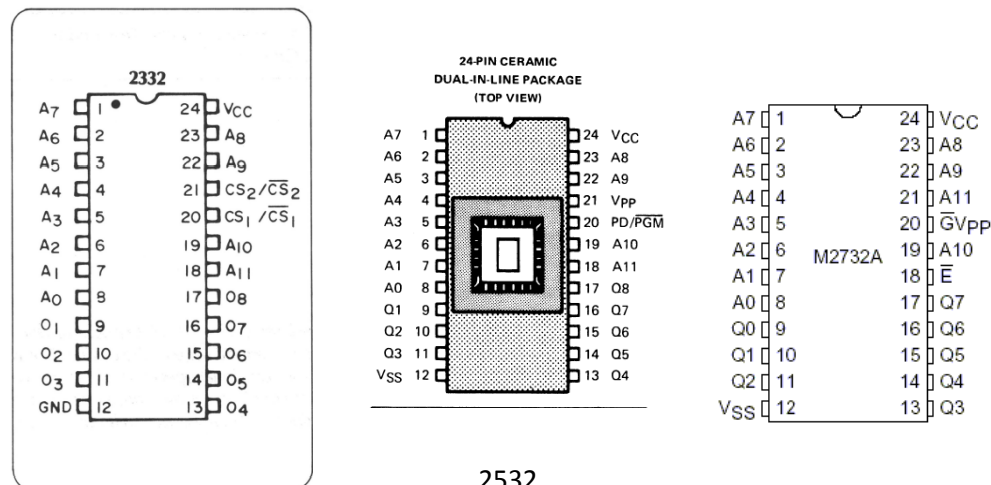
- Finish the adapter by soldering the socket for the EPROM on the solder side of the PCB.

Appendix A: pin outs of the devices involved

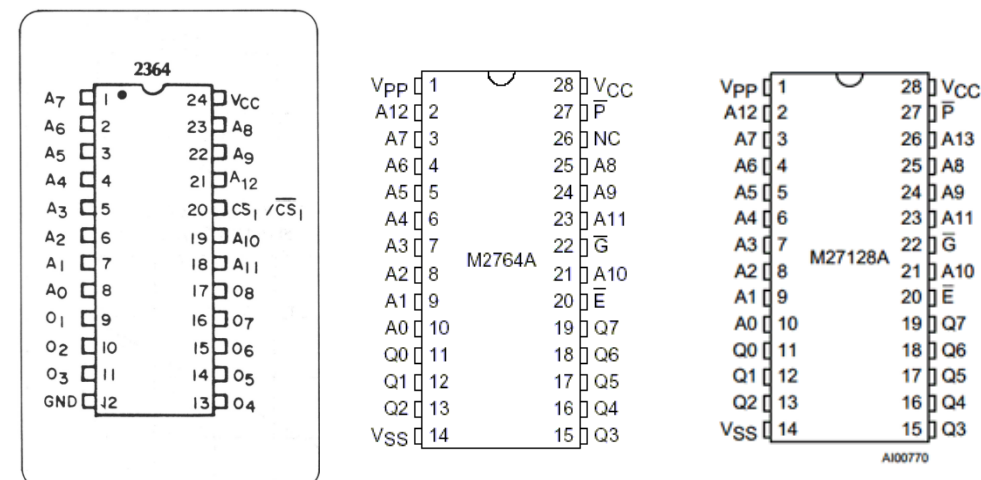
2316 and 2716



2332, 2532 and 2732



2364, 2764 and 27128



Note: some 2764's require pin 26 to be logic high for normal operation or offer the pin as an alternative for pin 28 as the supply pin.

Please note that EPROMs always have a CS pin that is active low, and apart from the 2532 have an output enable pin that is also active low. In addition, EPROMs have a pin that carries the programming voltage during programming; for read operation the pin is to be connected to Vcc or at least be at logic high. Note that on the 2732 Vpp is shared with /OE; when logic high, its outputs are disabled.

The 23XX series of ROMs have one or more chip select pins. The polarity of the pin(s) is a mask option and can be either active low or active high. In practice most ROMs used in Commodore equipment have been made with polarities that are compatible with the matching EPROM; the 2364's used in the Commodore 8050/8250 floppy disk drives being the notable exception.

Appendix B: bills of material

2332 footprint adapter

Ref.	Description	Value	Qty.	Remark
S1	Pin strip, 12 pins	-	2	Use two strips of 12 pins if you can get them, otherwise use two larger ones and cut to length.
U1	IC socket, 24 pin, 600 mil wide	-	1	
U2	74LS00	DIL	1	
C1,2	Capacitor, ceramic, 200mil spacing, radial	100nF, 16V or more	2	
T1,2	Angled jumper pin row, 2 pins	-	1	Optional: only needed for the function 2x2716 to 1x2732
-	PCB			

2364 footprint adapter

Ref.	Description	Value	Qty.	Remark
U1	Pin strip, 12 pins	-	2	Use two strips of 12 pins if you can get them, otherwise use two larger ones and cut to length.
U1A	IC socket, 28 pin, 600 mil wide	-	1	
U2	74LS00	SO package	1	
C1,2	Capacitor, ceramic, 200mil spacing, radial	100nF, 16V or more	2	
R1	Resistor, 1/4W, 0207	4k7ohm		
A12	Angled jumper pin row, 2 pins	-	1	Optional: only needed for the function 4x2332 to 1x27128. If used, top bridge of IC socket U1A needs to be removed.
A13	Straight jumper pin row, 2 pins	-	1	Always needed
CS POL	Angled jumper pin row, 2 pins			Optional
-	PCB			
